



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-1:2015CORE ABILITIES	
TAHAP / LEVEL	1 (ONE)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA04HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	4.1 ADHERE TO HEALTH AWARENESS ACTIVITY 4.2 ADHERE TO SAFETY AWARENESS ACTIVITY 4.3 ADHERE TO ENVIRONMENTAL AWARENESS ACTIVITY	
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TAJUK/TITLE: ENVIRONMENTAL AWARENESS CONCEPTS

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the Environmental Awareness Concepts. The person who is competent in this CA shall be able to adhere to health awareness activity, adhere to safety awareness activity and adhere to environment awareness activity.

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PENERANGAN/INFORMATION:

1. Environmental Awareness Activities

The environment is defined as 'living things, their physical, biological and social surroundings and the interactions between all of these. Environmental management measures will depend on the nature of the site activities and the sensitivity of the project area and surrounding land or water environment. For example, excavations resulting in steep slopes are likely to lead to soil erosion and water quality problems downstream and will require the installation of erosion protection measures.

Environmental pollution and ecological degradation has resulted in economic losses for many developing countries such as air pollution and water pollution. However, high economic growth has put immense pressure on the government to initiate and increase the environment protection and face challenges in combating environmental degradation.

Environmental education is defined as a process directed at creating awareness and understanding about environmental issues that leads to responsible individual and group actions. Successful environmental education focuses on processes that promote critical thinking, problem solving, and effective decision-making skills. Environmental education utilizes processes that involve individuals in observing, measuring, classifying, experimenting and data gathering techniques. These processes will assist them in discussing, inferring, predicting and interpreting data about environmental issues.

1.1 Current Environmental Issues

Engineering developments are resulting in resource depletion and environmental destruction. Modern technologies used in the engineering and manufacturing industry have a major impact on our life in past few years. Due to the rapid changes in the engineering and manufacturing industry have been drastic changes in the environment.

Engineering and manufacturing industry have increased the use of materials like metals, plastic, oil and rubber. These are used in the production of numerous end products which can be associated with different industries such as Car production units, shipping industries, Cotton mills, plastics

industries, Coal mining, heavy machineries and other which are causing numerous arduous effects and are considered to be non-environment friendly.

Crucial environmental issues are no more a blame game. While most of us crib about dirty air, smelly garbage or polluted water, least do we know it is “us” who is responsible for this unfavourable circumstances leading to cautionary environmental issues.

Here are 10 significant current environmental issues, where human beings play an important role in its cause.

- i. Pollution: More than half of the human population knows what pollution is, but we are still not ready to face its damaging consequences. Pollution is not only limited to water, soil and noise but has extended to light, visual, point and non-point sources. Human beings and their actions are majorly responsible for causing all types of pollution. Water pollution is essentially cause by oil spills, urban runoff and ocean dumping. Air pollution rises from burning of fossil fuels, hydraulic fracturing and gases emitted by vehicles. Water and soil pollution are majorly cause from industrial waste.



- ii. Climate Change: Climate change today is less of a natural process. It is rapidly occurring due to the ill effects of human actions responsible for disturbing and harmful out comings such as global warming, greenhouse effect, urban heat, coal industry etc. Climate change is not only changing the overall weather scenario, but has larger and harmful effects. Some of these include: melting of

polar regions, occurrence of new diseases and permanent inhibition in growth of certain plants essential for human survival.

- iii. **Global Warming:** Global warming is another environmental issue which is increase in earth's temperature due to effect of greenhouse gases called carbon dioxide, methane, water vapor and other gases. These gases possess heat trapping capacities that are needed to create greenhouse effect so that this planet remains warm for people to survive. Without these gases, this planet would turn be cold for life to exist. During past several decades, the accumulation of greenhouse gases have grown rapidly, which means more heat gets trapped in the atmosphere and few of these gases escapes back into the space. These gases heat up the earth's surface and this result in global warming. According to Environmental Protection Agency (EPA) reports, the earth's temperature has increased by 0.8 degrees Celsius over the past century. Global warming is a serious public health and environmental concern. Global warming can have long lasting effects which can result in melting of glaciers, climate change, droughts, diseases and increase in hurricanes frequency.
- iv. **Deforestation:** With population growing at a rapid pace, the demand for food, shelter and cloth has almost tripled in last few decades. To overcome growing demand, a direct action that we have come to recognize as "Deforestation" occurs. Deforestation means, clearing of forests or green cover for means of agriculture, industrial or urban use. It involves permanent end of forest cover to make that land available for residential, commercial or industrial purpose.
- v. **Overpopulation:** This is a never-ending human tragedy which is responsible for causing all types of environmental issues. Water pollution, resources crisis, gender imbalance, pollution, land pollution, urban sprawling, deforestation, over production are some common examples of dangerous effects cause by overpopulation. Despite efforts taken by the government in terms of family planning in many countries, over population is difficult to control at international level. This has become more like a subjective concern and no method seems to be 100% efficient to resolve the problem of over population.

- vi. Industrial and Household Waste: At present, tons of garbage is produced by each household each year. Items that can be recycled are sent to local recycling unit while other items become a part of the landfills or sent to third world countries. Due to increase in demand for food, shelter and house, more goods are produced. This resulted in creation of more waste that needs to be disposed of. Most waste is buried underground in landfill sites. The presence of huge landfills sites across the city pose serious environmental concerns. It affects human health, degrades soil quality, effects wildlife, cause air pollution and results in climate change.
- vii. Acid Rain: Acid rain simply means rain that is acidic in nature due to the presence of certain pollutants in the atmosphere. These pollutants come in the atmosphere due to car or industrial processes. Acid rain can occur in form of rain, snow, fog or dry material that settle to earth. Acid rain may cause due to erupting volcanoes, rotting vegetation and sea sprays that produce sulphur dioxide and fires, bacterial decomposition and lightening generate nitrogen dioxide.

Acid rain can also be caused due to man-made sources which include combustion of fossil fuels which release sulphur dioxide and nitrogen oxides into the atmosphere. Acid rain can have devastating effects on aquatic life, forests, public health and architecture and buildings.

- viii. Ozone Layer Depletion: Ozone layer is a layer of gas that sits 25-30 km above earth's surface. It mainly contains contain ozone which is a naturally occurring molecule containing three oxygen atoms. This layer is present in the stratosphere and prevents too many harmful UV (ultra violet) radiations from entering the earth. Ozone layer is capable of absorbing 97-99% of the harmful ultraviolet radiations that are emitted by sun.

However, during last several decades, human and industrial activity has contributed a lot which has resulted in considerable reduction in the ozone layer of the atmosphere. The main cause of depletion of ozone layer is determined as excessive release of chlorine and bromine from man-made compounds such as chlorofluorocarbons (CFCs). CFCs (chlorofluorocarbons), halons, CH₃CCl₃

(Methyl chloroform), CCl₄ (Carbon tetrachloride), HCFCs (hydro-chlorofluorocarbons), hydrobromofluorocarbons and methyl bromide are found to have direct impact on the depletion of the ozone layer.

- ix. Genetic Engineering: Genetic modification of food, human and animal organs seems like the gem of science and technology but this has major harmful effects. Biotechnology is an impressive technology but limiting its use is the need of the hour. Genetic engineering is a controversial subject and has seen more ill impacts than the benefits it brings to mankind. Genetic pollution and alteration of food produce not only have harmful effects on human beings, but are responsible for crucial concern known as 'genetic modification'.
- x. Urban Sprawl: Not only India and China are classic examples of over population and urban sprawl leading to land degradation. Today almost all countries are using the land irresponsibly to meet the ever-growing demand of the greedy human wishes. The expansion of industrial areas has not only led to land degradation and soil pollution, but the habitat destruction is a terrible misery. Natural environment consisting of flora and fauna is indiscriminately destroyed and lost completely instead of being replaced. This in the long run has harmful impact for human survival and cause serious environmental issue.

2 Environmental Control Act And Regulations

In industrial countries, voluntary environmental agreements often provide a platform for companies to be recognized for moving beyond the minimum regulatory standards and thus support the development of best environmental practice.

In developing countries, such agreements are more commonly used for remedy significant levels of non-compliance with legislations and regulations. The key advantage of its use in developing countries is to build environmental management capacity. The legislation that is related to the prevention, abatement, control of pollution and enhancement of the environment in Malaysia is the Environmental Quality Act, 1974. PART IV of the Environmental Quality Act 1974 (Act 127), is referred to for prohibition and control of environmental pollution.

3 Signage And Definition

Signage system is referred to a group of signs such as symbols, maps, arrows, colour codings, pictograms and different typographic elements with coherent designs and purposes. Normally significant efforts are set into creating an intelligent presentation for a sign that takes into account scientific knowledge on human reactive capabilities. Some of the most common signage used to indicate environmental related commands are as follows.

	Sign indicating campaign on conserving the environment by keeping the nature ecology of the earth cleaned.
	A sign to encourage members of public to save the earth by recycling the use of environment destructive substances.
	A signage on reducing of water consumption wastages in effort to conserve the environment and maintain resources for the future.
	Signage in effort to keep the environment cleaned and awareness among members of public in rubbish dumping practice.

**Going
Greener**

A signage to encourage members of public in keeping sustainability of the earth by selecting correct bin, identified by its colour, for waste disposal. The disposed items are meant for recycling purpose.

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SOALAN/QUESTION :

1. State the nature of economic losses in developing countries attributed by environmental pollution and ecological degradation? - air pollution and water pollution
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2. State a reason why the government initiates the environment protection initiatives? – high economic growth in the country
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3. What environmental education? - a process directed at creating awareness and understanding about environmental issues that leads to responsible individual and group actions.
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4. How to achieve a successful environmental education? - focuses on processes that promote critical thinking, problem solving, and effective decision-making skills.
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5. Choose a correct technique to achieve environmental education process? - observing, measuring, classifying, experimenting and data gathering techniques.
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6. These processes will assist them in discussing, inferring, predicting and interpreting data about environmental issues.
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7. How do voluntary environmental agreements function in industrial countries? - provide a platform for companies to be recognized for moving beyond the minimum regulatory standards and thus support the development of best environmental practice.

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8. How do voluntary environmental agreements function in developing countries? - commonly used for remedy significant levels of non-compliance with legislations and regulations.

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9. What is the key advantage of voluntary environmental agreements in developing countries? - to build environmental management capacity.

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10. State the legislative document that covers the enforcement of environmental quality in Malaysia. - Environmental Quality Act 1974 (Act 127)

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11. Which part of Environmental Quality Act 1974 (Act 127) that refers to prohibition and control of environmental pollution. – PART IV.

RUJUKAN / REFERENCES:

1. Environmental Quality Act 1974 (Act 127): regulations, rules & orders,(2009) ISBN 978-967891-934-0
2. Occupational Safety and Health Act 1994 and Regulation, (2009) Malaysia, Penerbitan Akta (M), ISBN-967701188-X.
3. Department of Environment, Ministry of Science, Technology and Environment (2000). “A Handbook of Environmental Impact Assessment Guideline.”: Department of Environment, Malaysia.